

SWATI YADAV

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EDUCATION

National Centre for Biological Sciences- TIFR

PhD, Biology

Bangalore

2017 - 2024

University of Hyderabad

Integrated BS-MS, Chemical Sciences

Hyderabad

2012 - 2017

TECHNICAL SKILLS

Membrane protein biochemistry: Membrane protein expression, Protein purification, Activity assay, Biochemical analysis, Native PAGE, Lipid nanodisc reconstitution, Liposome reconstitution, Lipid TLC

Electron microscopy: Negative stain EM, Cryo-EM sample preparation, Data collection and processing using RELION and cryosparc, Model building using Coot, PHENIX, Refmac Servalcat

Cell culture: Protein over-expression, Baculovirus generation, BacMam, Bacteria culture, Yeast culture, Mammalian cell culture, Sf9 cell culture

Molecular biology: Sub-cloning, Site directed mutagenesis

PUBLICATIONS

- **Yadav, S.,** & Vinothkumar, K. R. (2024). Factors affecting macromolecule orientations in thin films formed in cryo-EM. *Acta Crystallographica Section D*, 80(7), 535–550. <https://doi.org/10.1107/S2059798324005229>
- **Yadav, S.,** Jha, S., Vinothkumar, K. R. (2024), Chapter 22, Cryo-EM of mitochondrial membrane proteins, *Cryo-Electron Microscopy in Structural Biology: From Structural Insights to Tomography and Drug Discovery* (1st ed.). CRC Press. <https://doi.org/10.1201/9781003326106>
- **Yadav, S.,** Banik, S., Prasad, M. D. (2021). Understanding of the C-H stretch region of infra-red spectroscopy: an analysis of the final state wavefunctions. *Physical Chemistry Chemical Physics*, 23(15), 9176–9188. <https://doi.org/10.1039/d0cp01157f>
- Das, A., Vishwakarma, V., Dey, A., Dey, S., Gupta, A., Das, M., Vishwakarma, K. K., Roy, D. S., **Yadav, S.** Kesarwani, S., Venkatramani, R., Maiti, S. (2021). Biophysical properties of the isolated spike protein binding helix of human ACE2. *Biophysical Journal*, 120(14), 2785–2792. <https://doi.org/10.1016/j.bpj.2021.06.017>
- Arya, C. K., **Yadav, S.,** Fine, J., Casanal, A., Chopra, G., Ramanathan, G., ... Subramanian, R. (2020). A 2-Tyr-1-carboxylate Mononuclear Iron Center Forms the Active Site of a Paracoccus Dimethylformamidase. *Angewandte Chemie - International Edition*, 59(39), 16961–16966. <https://doi.org/10.1002/anie.202005332>

RESEARCH EXPERIENCE

Post doctoral fellow

National Centre for Biological Sciences, Bangalore, Karnataka, India

Sep 2024- current

- Mechanosensitive channel, Restriction enzyme complex

Research Associate

National Centre for Biological Sciences, Bangalore, Karnataka, India

Nov 2023- Sep 2024

- Mechanosensitive channel, Restriction enzyme complex

Graduate student

National Centre for Biological Sciences, Bangalore, Karnataka, India

Aug 2017 - Nov 2023

- Cryo-EM, Preferred orientation problem, Membrane protein biology

Undergraduate thesis project student

University of Hyderabad, Hyderabad, Telangana, India

Jan 2017 - May 2017

- Computational chemistry, IR spectroscopy

Summer Intern

National Chemical Laboratory, Pune, Maharashtra, India

May 2016 - Jul 2016

- Dye sensitized solar cells, Catalysis

SELECTED PROJECTS

- **Open state structure of a chloroplast mechanosensitive channel**

Investigated how membrane lipid composition regulates gating of a chloroplast mechanosensitive channel. Expressed and purified the protein in HEK293F cells using baculovirus expression. Reconstituted the channel into detergents, nanodiscs, and liposomes, and imaged using cryo-EM. Determined multiple open-state structures in distinct lipid environments. (Unpublished)

- **Mechanism of action of the type III restriction enzyme complex**

Studied how the type III restriction enzyme encodes sequence specificity, undergoes nucleotide-dependent conformational switching, and initiates long-range communication along DNA. Captured multiple cryo-EM structures of the complex with and without DNA, revealing early conformational states involved in DNA processing. Overcame severe orientation bias to optimize sample preparation. (Manuscript in preparation)

- **The factors affecting macromolecule orientation in single particle cryo-EM**

Characterized the causes of orientation bias in thin films formed in cryo-EM using standard samples. Analyzed and compared sample behaviour and orientation distributions by varying grid preparation conditions. (Published)

- **Understanding the C-H stretch region of IR spectroscopy**

Analyzed the C-H stretch region of the infrared spectrum of select small molecules using computational methods. (Published)

CONFERENCES AND WORKSHOPS

- **Instructor** - Workshop on Transmission Electron Microscopy, April 2025
- **Speaker** - Workshop on Transmission Electron Microscopy, April 2025
- **Speaker** - NCBS Annual Talks, January 2025
- **Poster presentation** - NCBS Annual talks, January 2024
- **Poster presentation** - GRC meeting on ion channels, July 2024
- **Poster presentation** - NCBS Annual talks, January 2024
- **Volunteer** - 45th Indian Biophysical society meeting, March 2023
- **Speaker** - EMBO practical course on Image Processing for Cryo-Electron Microscopy, September 2023
- **Poster presentation** - NCBS Annual talks, January 2021
- **Participant** - Asia Pacific Microscopy Conference, February 2020
- **Participant** - Structure Assisted Development of Novel Therapeutics Conference, February 2019
- **Poster presentation** - NCBS Annual talks, January 2018
- **Participant** - ACS Chemfest, January 2015
- **Participant** - Vijyoshi, December 2012

REFERENCES

1. **Dr. Vinothkumar Kutti Ragunath**, NCBS - TIFR, GKVK Post Bellary road, Bangalore - 560065, India
Email ID : vkumar@ncbs.res.in
2. **Dr. Ramaswamy Subramanian**, Bindley Bioscience Center, Purdue University West Lafayette, IN 47907 USA
Email ID : subram68@purdue.edu